

## **Create a small full-stack app for managing tasks.**

### **Requirements**

- Create tasks with title, description, status, and tags
- View all tasks
- Update task status
- Track activities for task creation and status updates
- Show latest activities in UI
- Expose a stats API for total tasks, count by status, and most common tag
- Use WebSocket so that task and activity changes appear instantly across connected clients

### **Tech**

- React
- Node.js + Express
- MongoDB
- Socket.IO or WebSocket

### **Time Limit**

60 minutes

### **Notes**

- Focus on functionality and code structure over UI polish
- Minimal UI is acceptable
- No auth required

# Features

## 1. Create Task

A user can create a task with:

- `title`
- `description`
- `status` (`todo`, `in-progress`, `done`)
- `tags` (array of strings)

## 2. View Tasks

Show all tasks grouped by status or in a simple list.

## 3. Update Task Status

A user can change the status of a task.

## 4. Real-Time Updates using WebSocket

When one client creates or updates a task:

- all connected clients should receive the update instantly
- UI should refresh without manual reload

## 5. Activity Feed

Track task actions:

- task created
- task status changed

Each activity should contain:

- `taskId`
- `action`
- `timestamp`

Show latest activities in the UI.

## 6. Basic Stats API

Return:

- total tasks
- count by status
- most common tag

Its okay if you do not want to display stats in frontend but keep backend stats API

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# WebSocket Requirements

Use WebSocket for these events:

## Emitted from backend

- `taskCreated`
- `taskUpdated`
- `activityCreated`

## Expected behavior

When a task is created or updated through the API:

- save in MongoDB
- create activity entry
- emit socket event to all connected clients

Frontend should listen and update UI accordingly.